

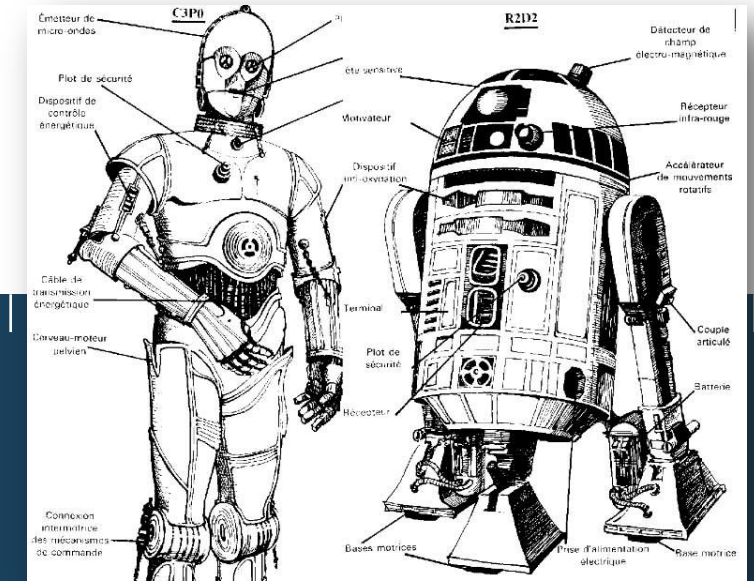




POLITECNICO
MILANO 1863



POLITECNICO
MILANO 1863



Robotics

Introduction

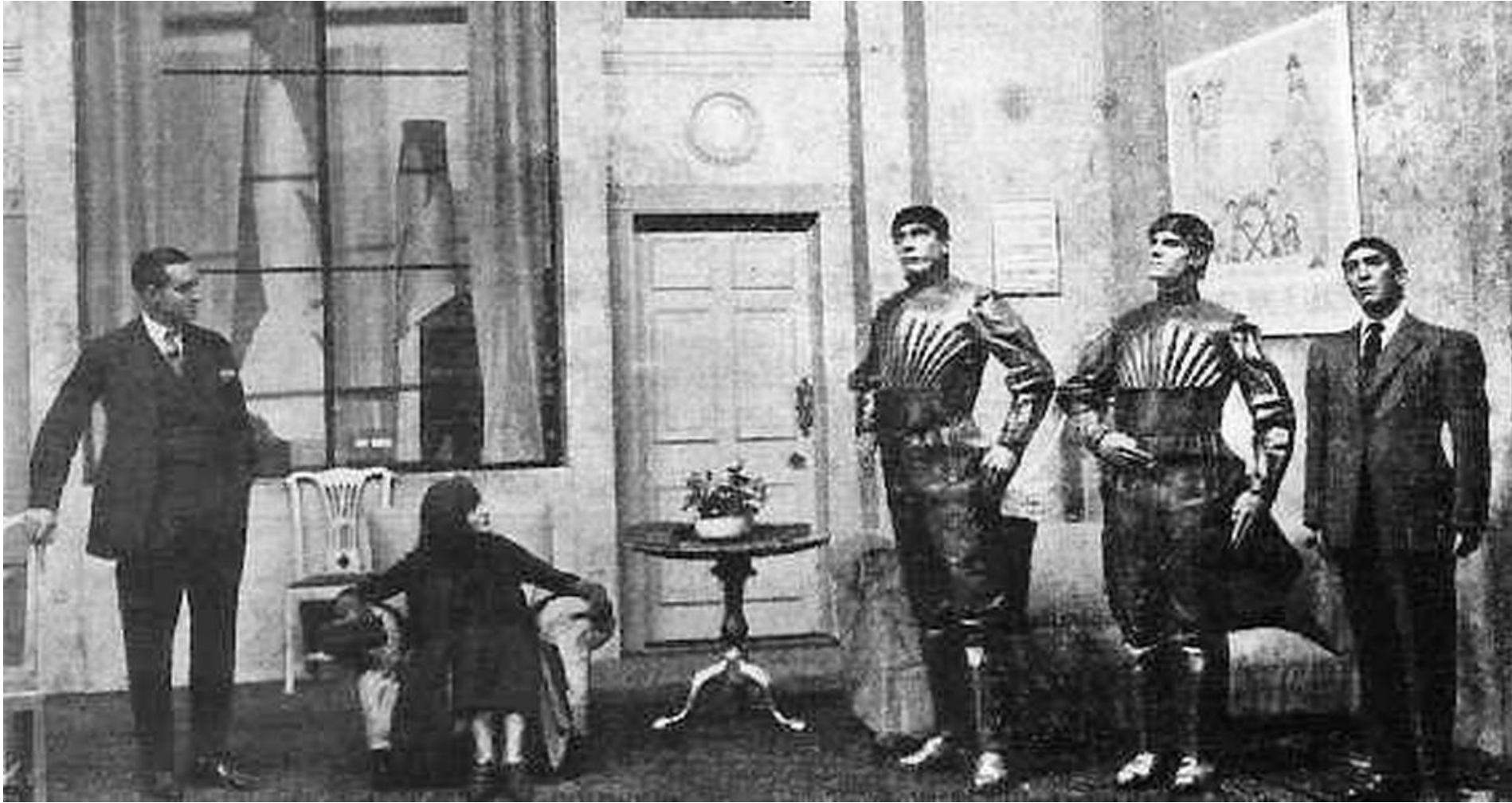
↙
"Robots" come from a 1920 play →

Matteo Matteucci
matteo.matteucci@polimi.it

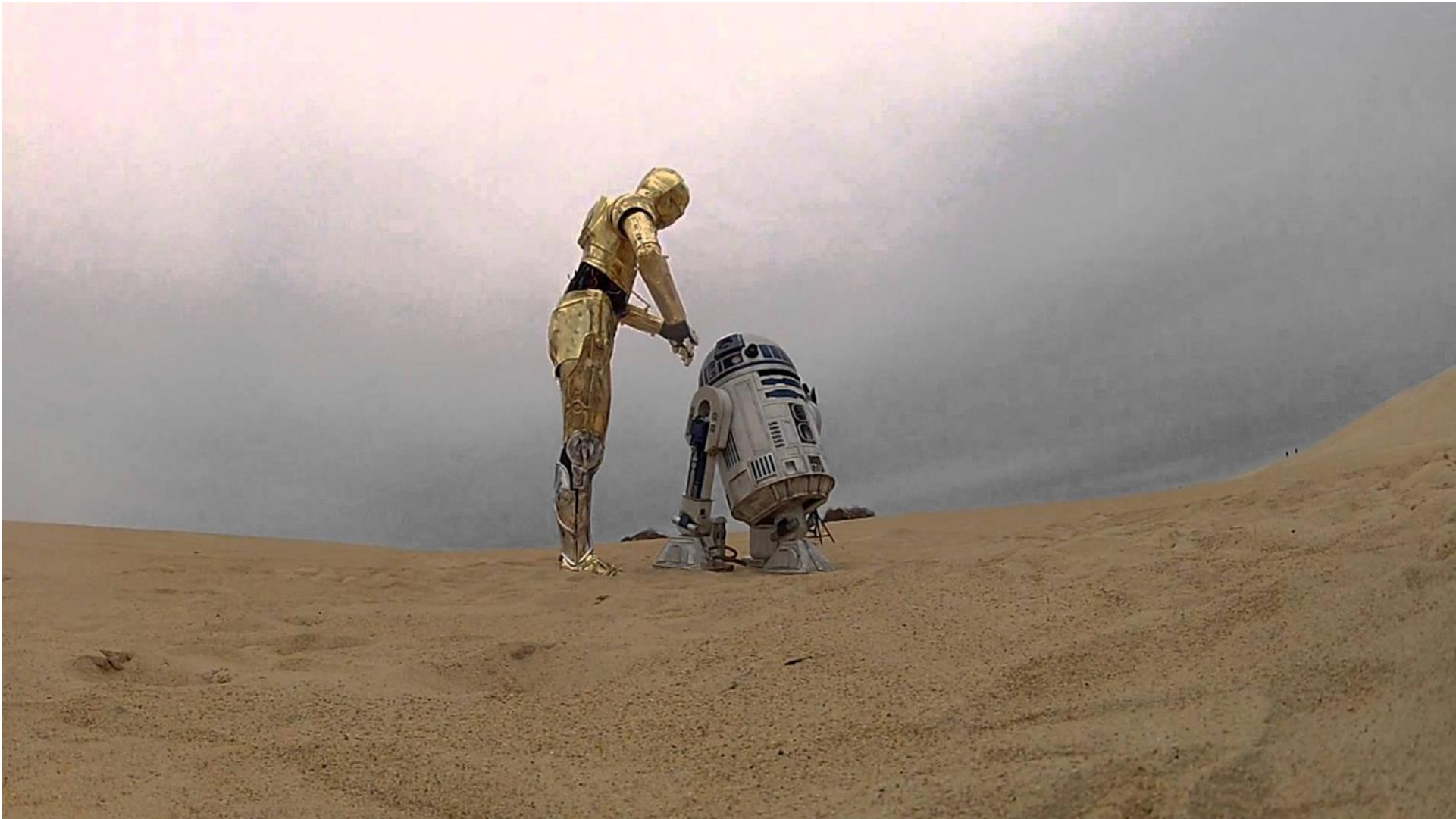
Artificial Intelligence and Robotics Lab - Politecnico di Milano

Rossum Universal Robots (1920)

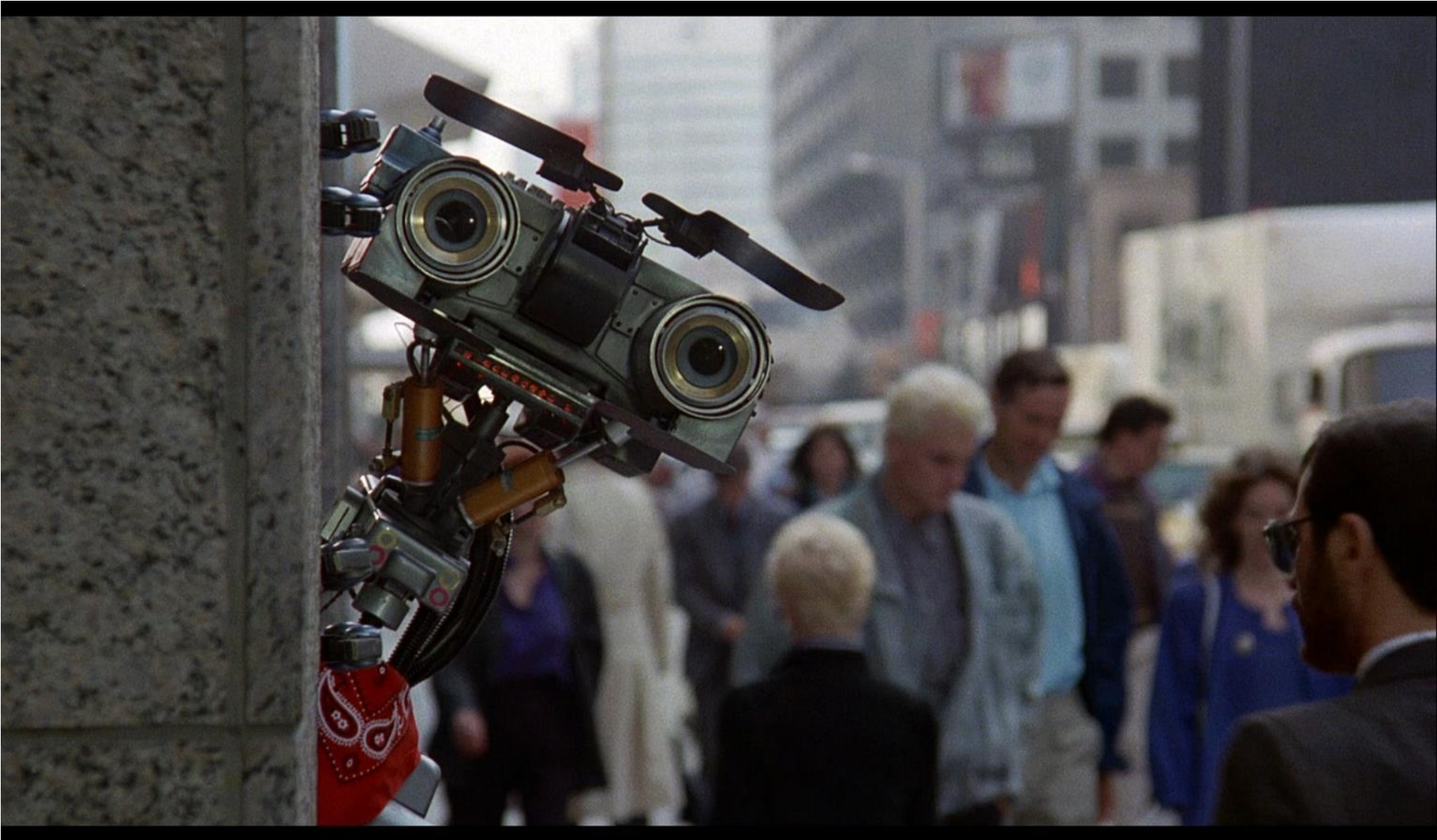
engineer
name of
the play
which
gave the
name
"ROBOT"



Star Wars (1977)



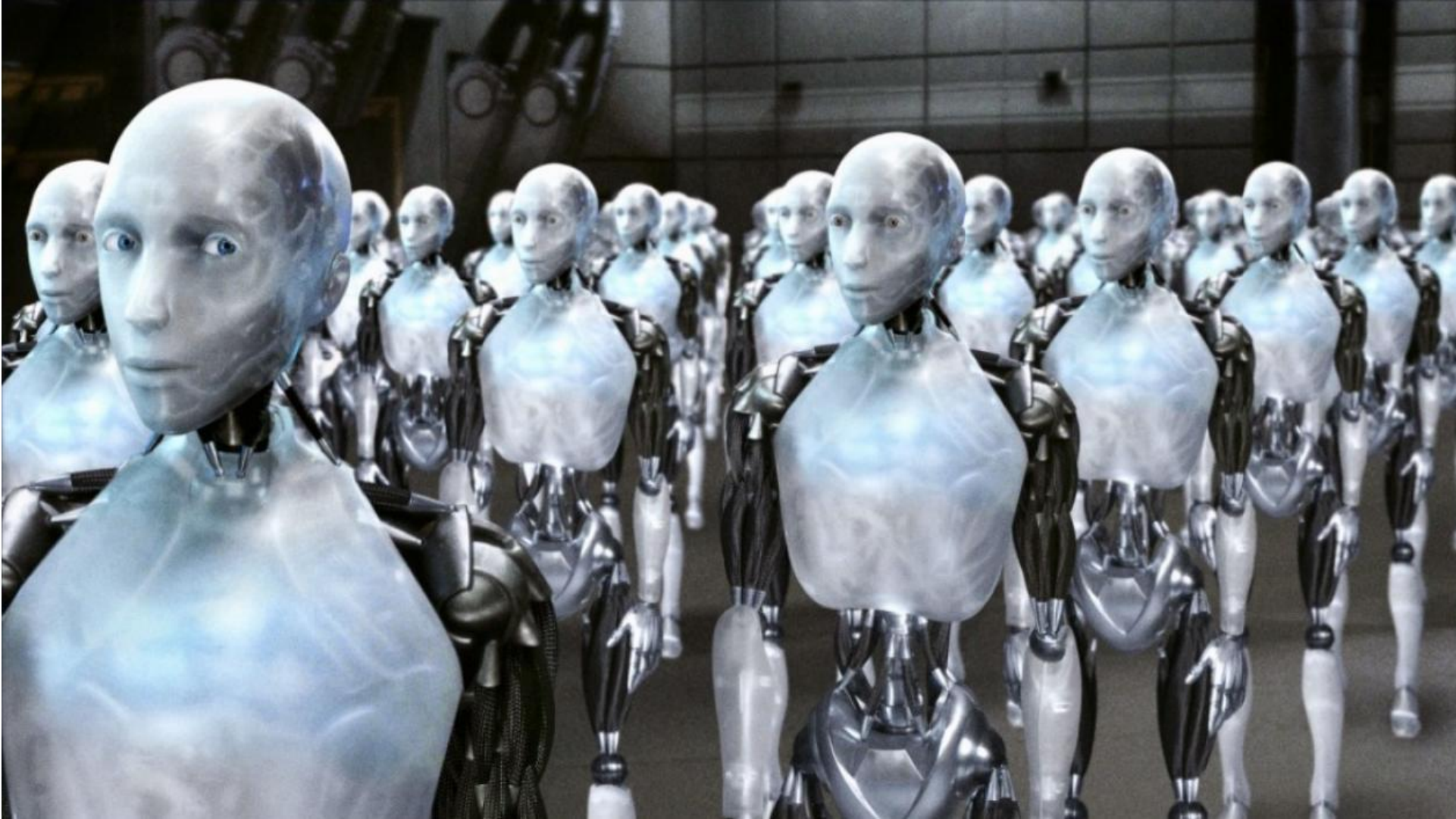
Short Circuit (1986)



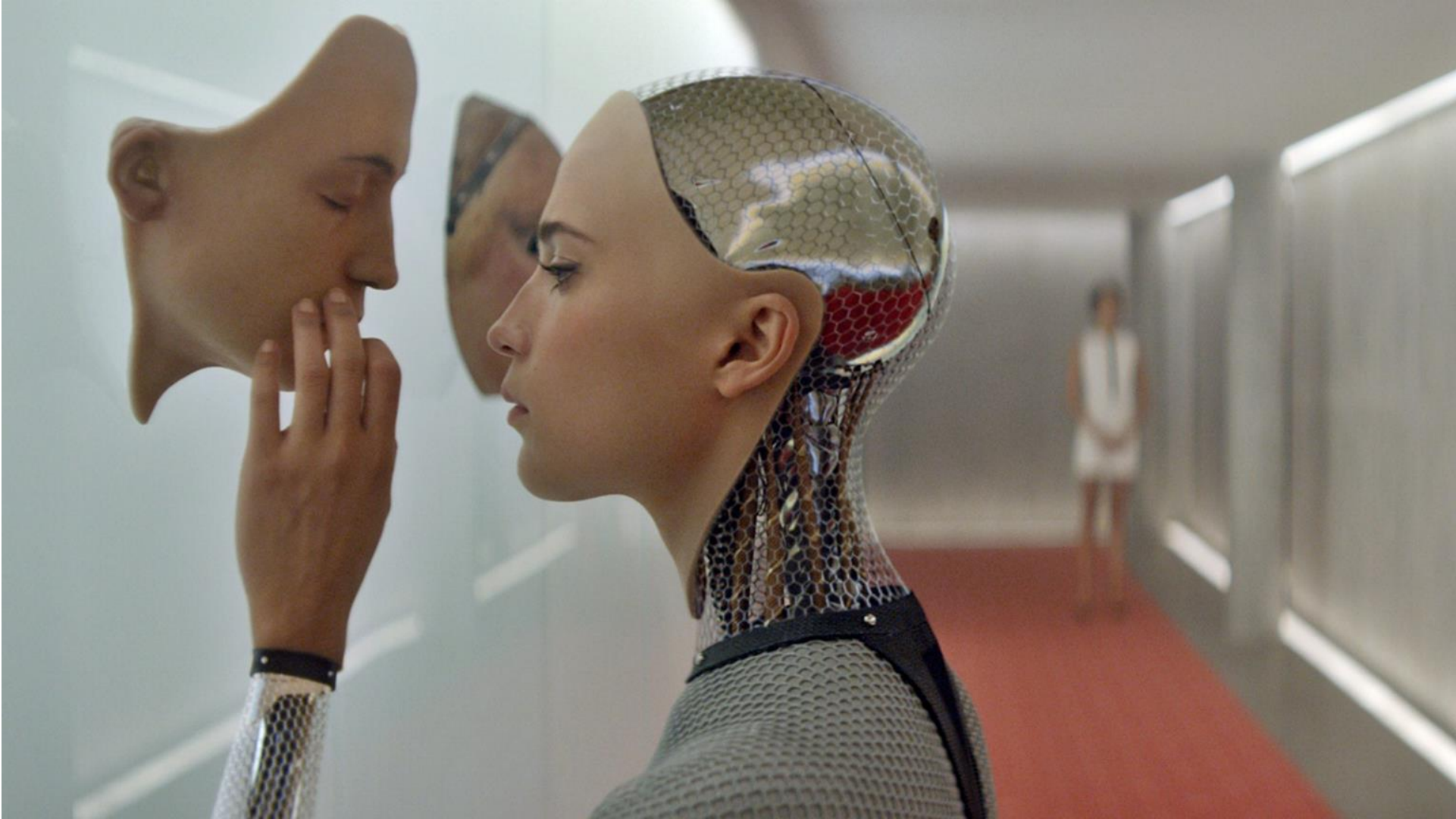
I Robot (2001) (humanoid Robots)

↓
from Asimov
books it
talks for
first time
about
"ROBOTICS"

↑
coming
from
sci-fi



Ex Machina (2015)

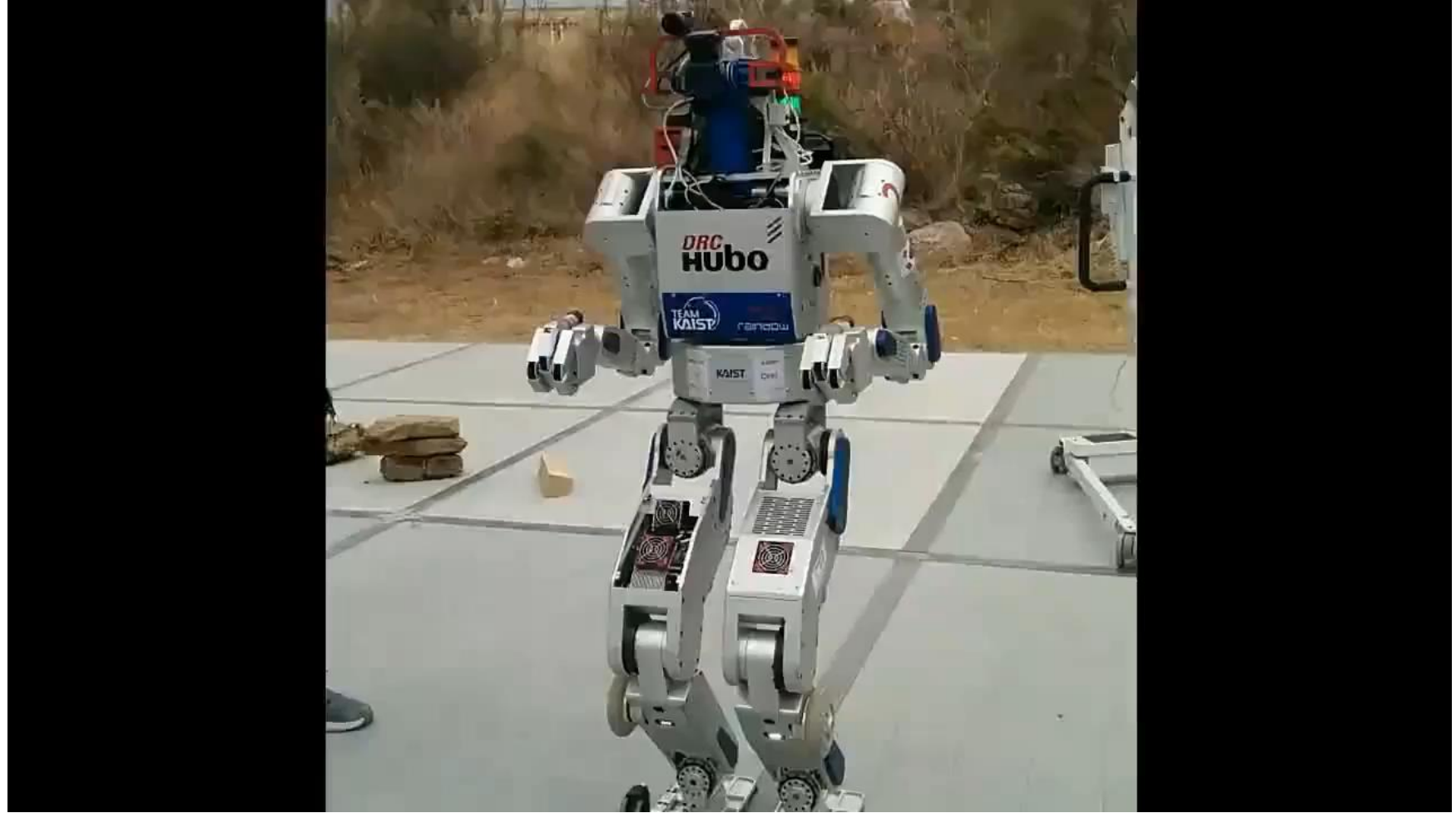
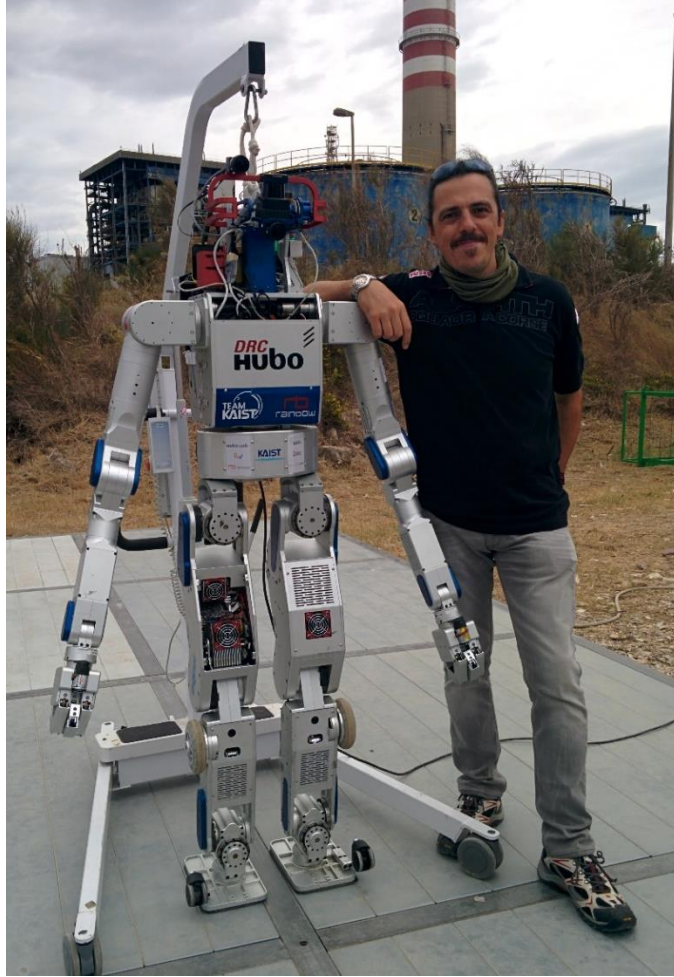


Sometimes reality is different...

(DRC)
↓
DARPA
Challenge
for human
robot
selection



... and the winner is ...



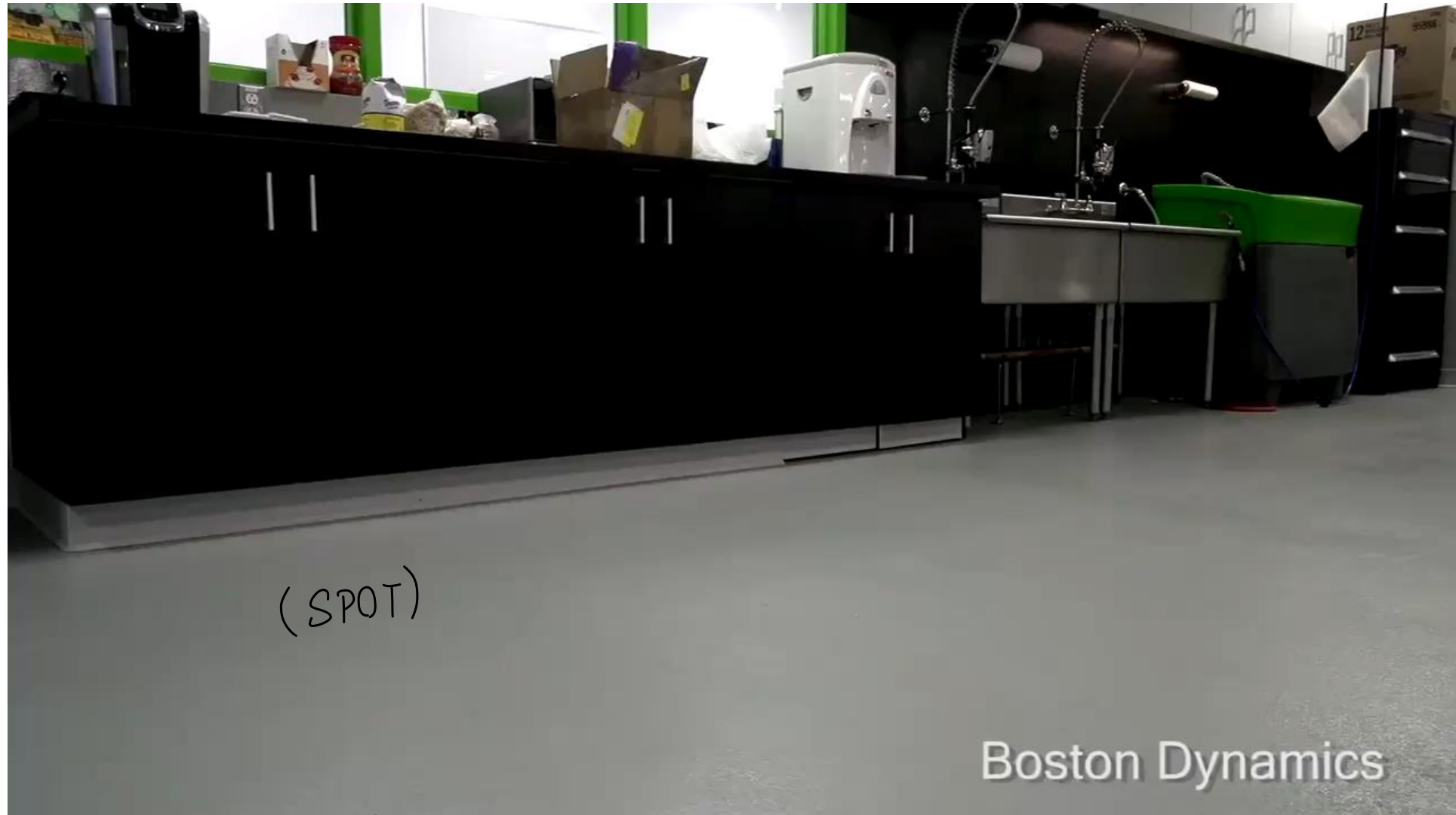
... and check! Sometimes dreams come true (**ATLAS** 2016) ...



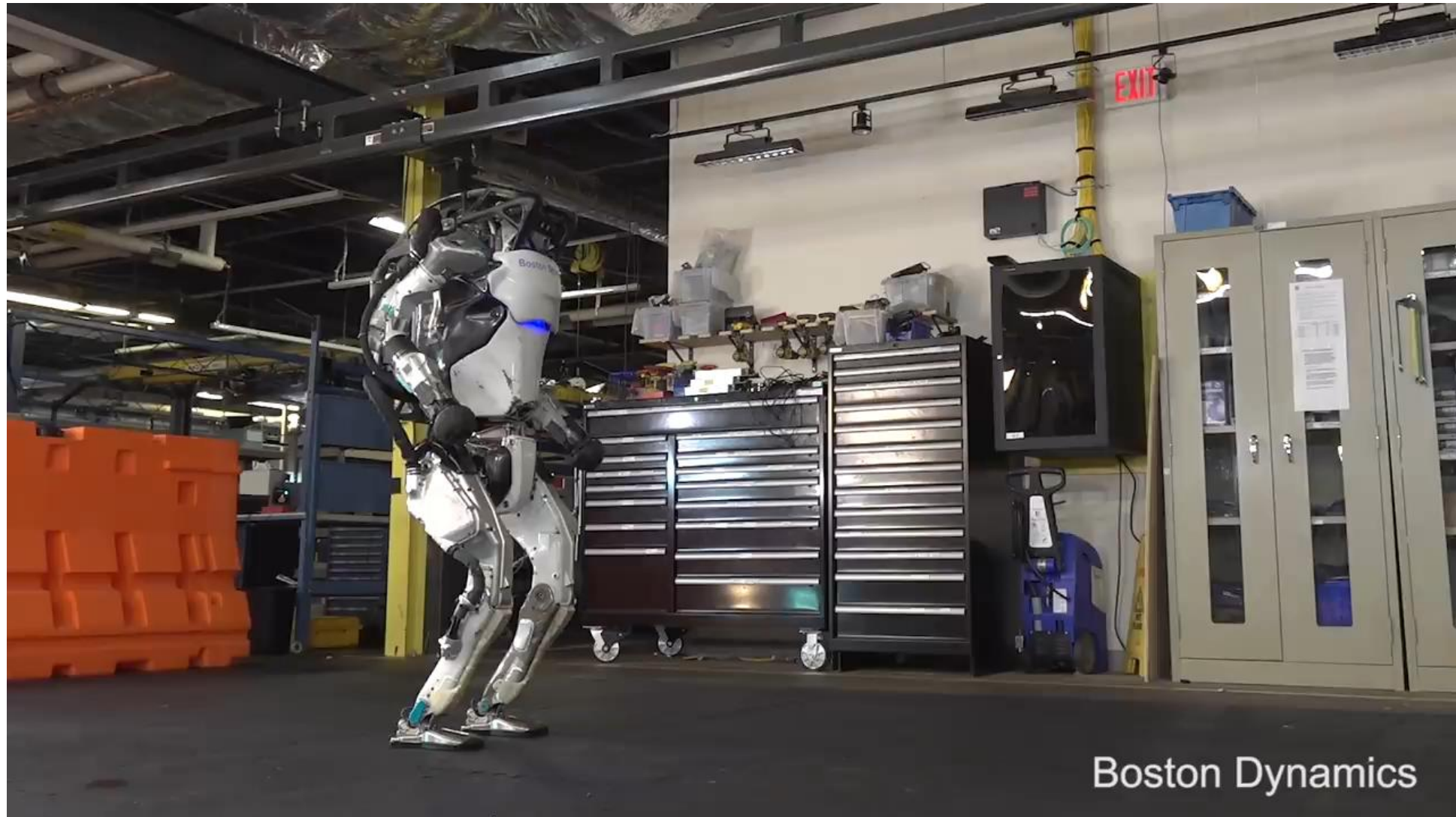
... and every year it gets better 😊



... and better 😊



... and better 😊



Steps in robot history → first real Robot in XVIII Century

Mechanical era (1700):

- automata
- karakuri-ningyo

by
mechanical
robots



Automata: the robot ancestors

mechanical systems

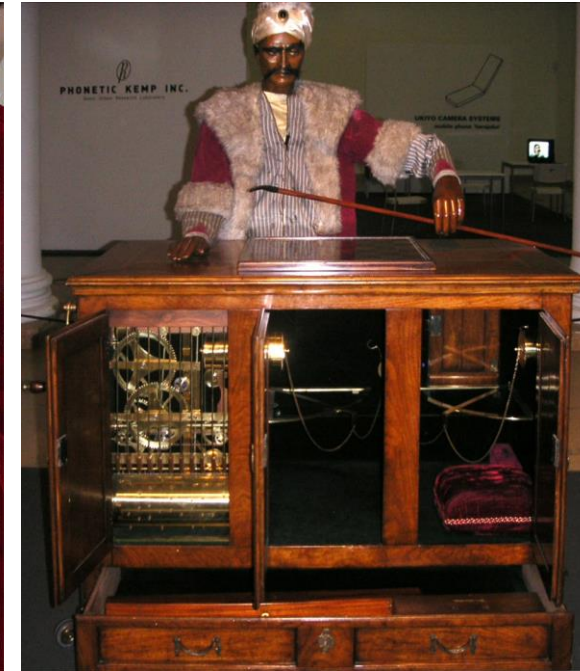
↑



Karakuri-ningyo
Edo Period
(1603 – 1868)



The Writer
Pierre Jaquet-Droz
(1721-1790)



The Turk
Wolfgang von Kempelen
(1734 – 1804)

Steps in robot history

Mechanical era (1700):

- automata
- karakuri-ningyo

firstly was
pure
mechanics



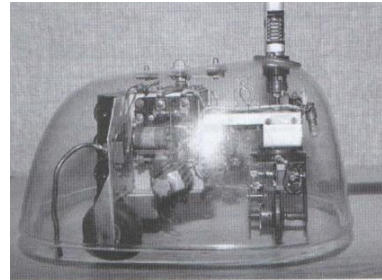
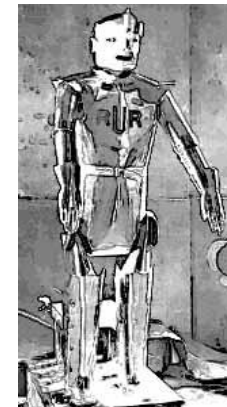
Fiction era ('20s):

- *Rossum Universal Robot*

Cybernetics era ('40s):

- Turtle and telerobot

mech
+ electronics



Early «robot» experiments

programmed as autonomous
robot experiment → ('40s for testing)

↑
Grey Walter's **tortoises** ('50s):
mechanical plausibility of animal
tropism. (Tropism/taxis: animal
movement directed by stimuli).



Steps in robot history

Mechanical era (1700):

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- karakuri-ningyo

Fiction era ('20s):

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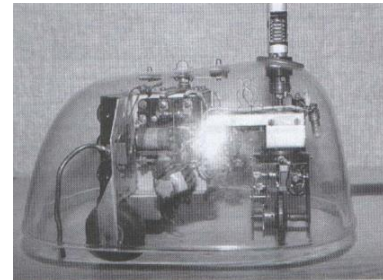
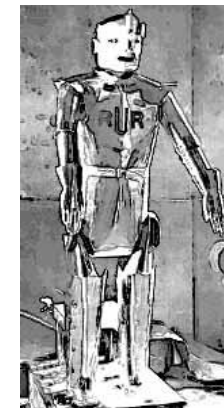
Cybernetics era ('40s): → research / pleasure

- Turtle and telerobot

Automation era (from the '60s):

- Industrial robots

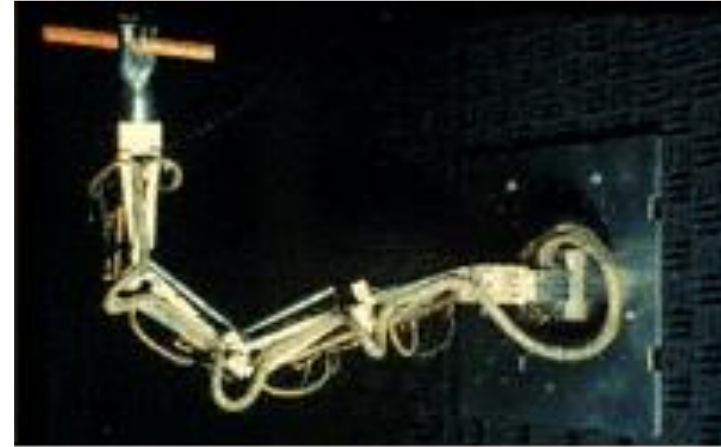
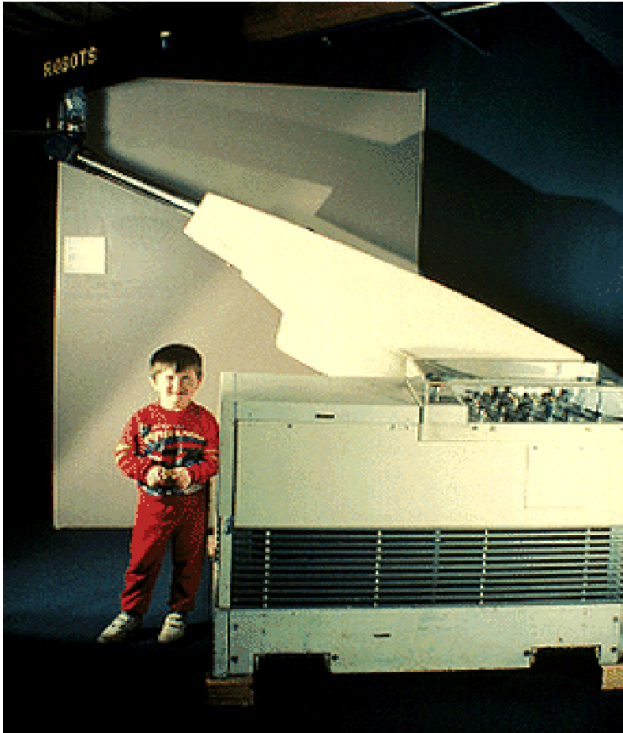
↑
Real Robot
used in
manufacturing



First robots

for moving big obj

1961 - **UNIMATE**, the first industrial robot, began work at General Motors. Obeying step-by-step commands stored on a magnetic drum, the 4,000-pound arm sequenced and stacked hot pieces of die-cast metal.



1968 - Marvin Minsky developed the **Tentacle Arm**, which moved like an octopus. It had twelve joints designed to reach around obstacles. A PDP-6 computer controlled the arm, powered by hydraulic fluids. Mounted on a wall, it could lift the weight of a person.

(Research)

What is a Robot?

↓ FIRST definition

A reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks.

(Robot Institute of America, 1980)

until '80
only
manipulators
↓
Robots
definition



An what about these???

We need a different
defintion of robot!

{ new
types
of Robots } ⇒



STARSHIP



Steps in robot history

Mechanical era (1700):

- automata
- karakuri-ningyo

Fiction era ('20s):

- *Rossum Universal Robot*

Cybernetics era ('40s):

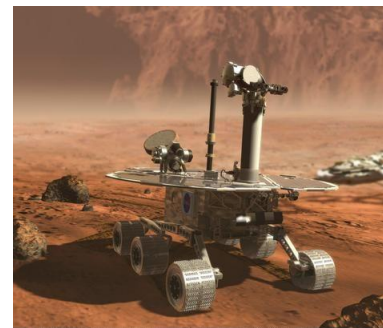
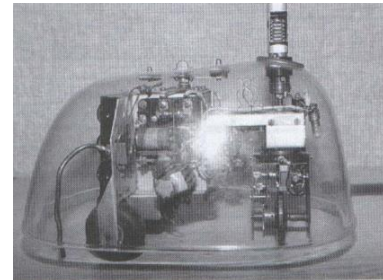
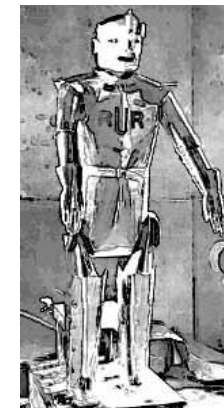
- Turtle and telerobot

Automation era (from the '60s):

- Industrial robots

Information era (from the '90s): $\Rightarrow$

- Intelligence
 - Autonomy
 - Cooperation
- $\hookrightarrow$ new robots with lot computy sw, perception not only mech
 $\hookrightarrow$ b/w multiple Rob.



Shakey (1972) and the Stanford Cart (1970)

↓ autonomous Robot



(MIT)

Indoor

{ to test autonomy of }
Robot



Outdoor



↓ new def of Robots

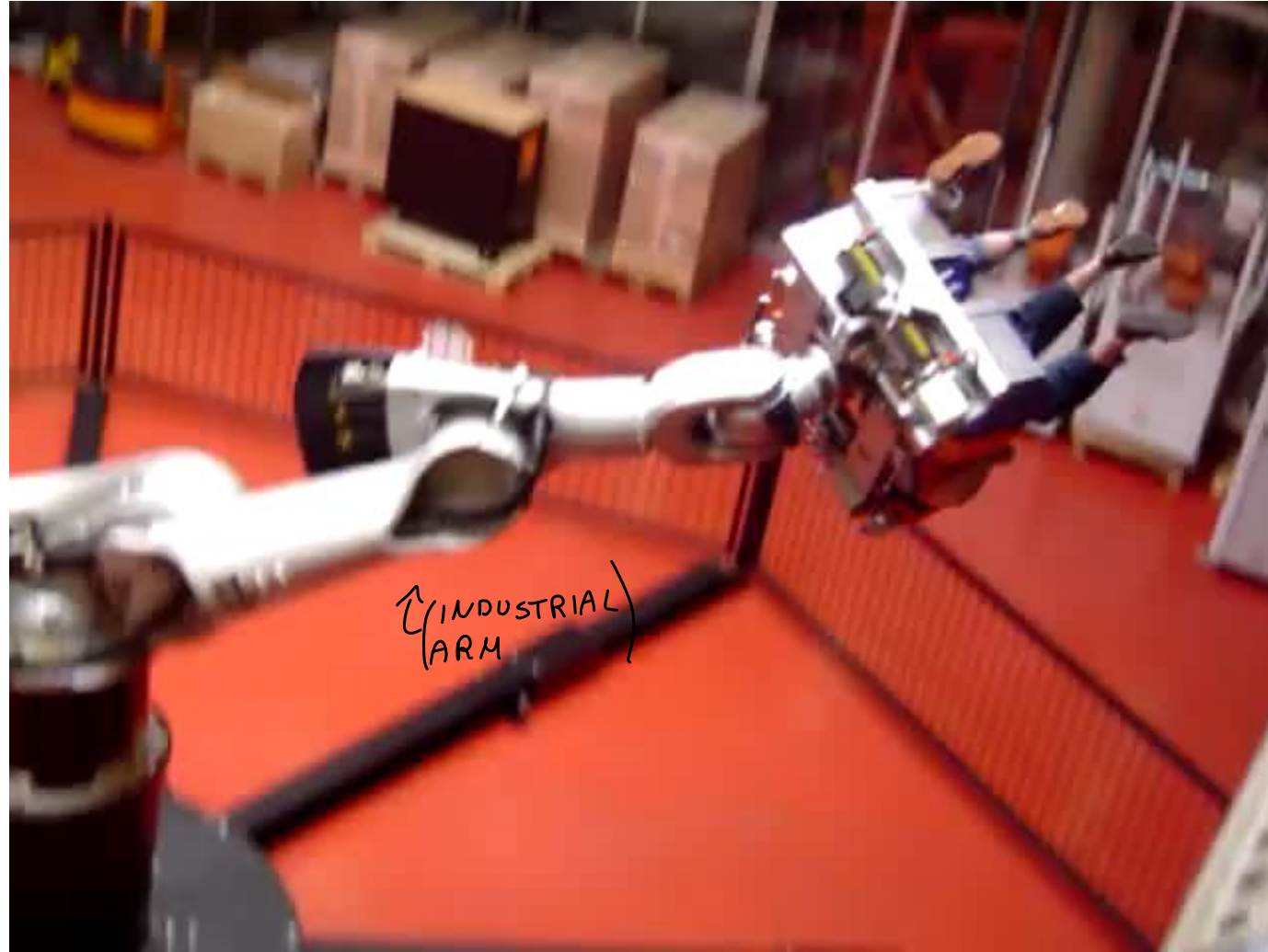
- ✓ A robot is an actuated mechanism programmable in two or more axes with a degree of autonomy, moving within its environment, to perform intended tasks. Autonomy in this context means the ability to perform intended tasks based on current state and sensing, without human intervention.
- ✓ A service robot is a robot that performs useful tasks for humans or equipment excluding industrial automation application.

Some operation
done without human help → some AUTONOMOUS

↓
to know the
context where the
Robot is working on

⇓
how to extract
info from sensors

Industrial vs Service Robotics



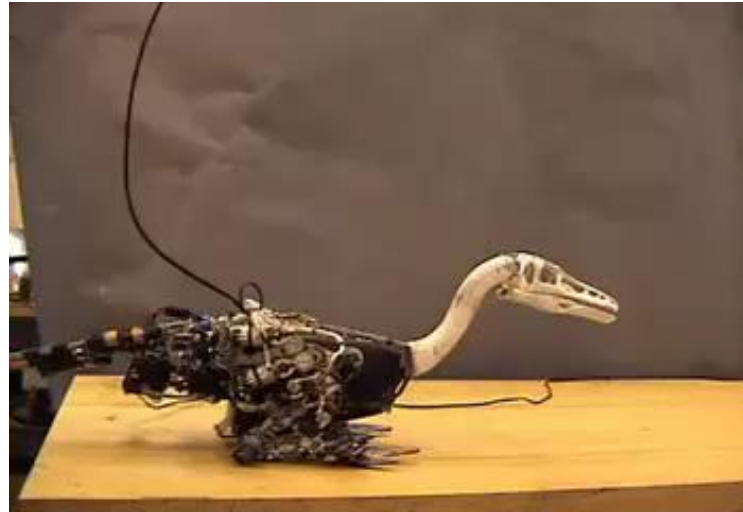
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- ✓ A **personal service robot** or a service robot for **personal use** is a service robot used for a **non-commercial task**, usually by lay persons. E.g., **domestic servant robot**, **automated wheelchair**, **personal mobility assist robot**, and pet **exercising robot**.

↑ to work with a human

Autonomous service robot

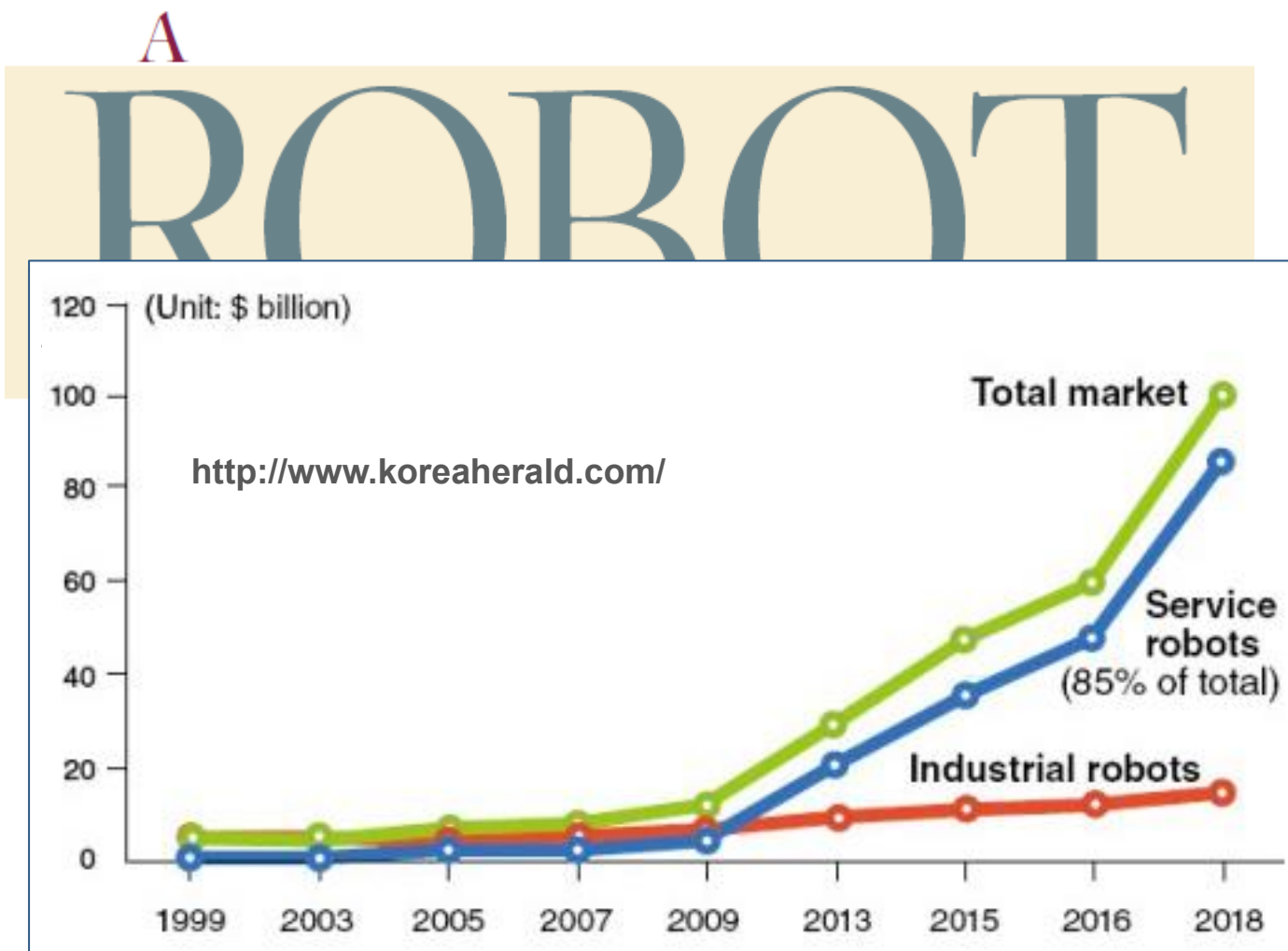


(Big dog)
by BOSTON
DYN



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- ✓ A personal service robot or a service robot for personal use is a service robot used for a non-commercial task, usually by lay persons. E.g., domestic servant robot, automated wheelchair, personal mobility assist robot, and pet exercising robot.
- ✓ A **professional service robot** or a service robot for professional use is a service robot used for a commercial task, usually operated by a properly trained operator. E.g., **cleaning robot** for public places, delivery robot in offices or **hospitals**, fire-fighting robot, rehabilitation robot and **surgery robot in hospitals**. In this context an operator is a person designated to start, monitor and stop the intended operation of a robot or a robot system.

The Bill Gates “Prophecy” ...

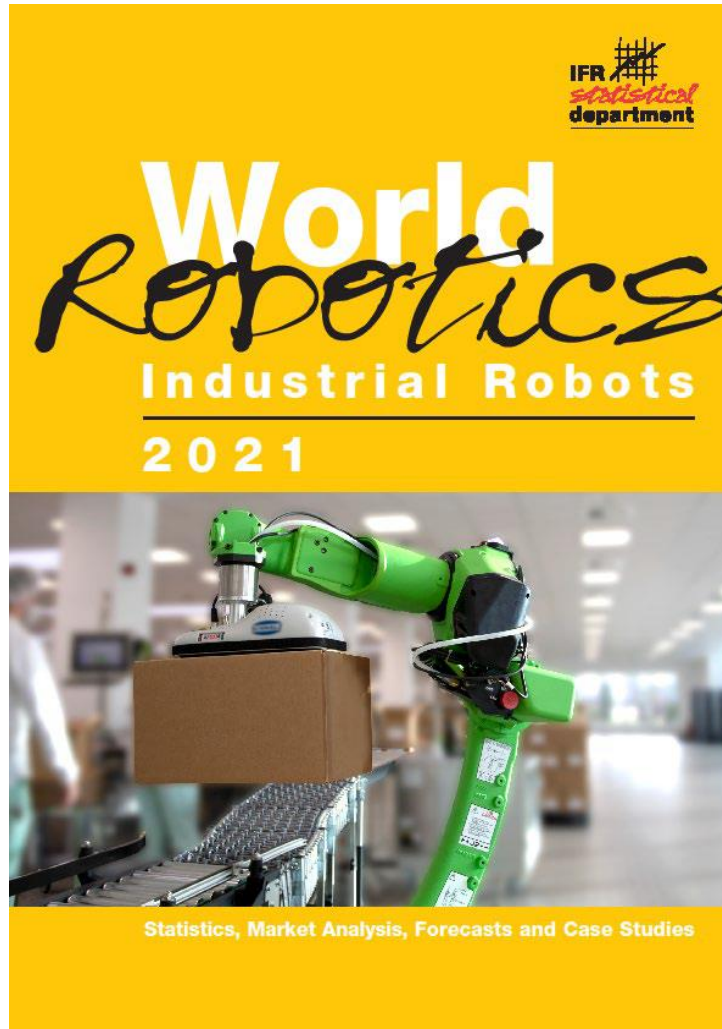


companies produce innovative toys, gadgets for hobbyists and other interesting niche products. But it is also a highly fragmented industry with few common standards or platforms. Projects are complex, progress is slow, and practical applications are relatively rare. In fact, for all the excitement and promise, no one can say with any certainty when—or even if—this industry will achieve



Industrial and Service Robots Market (<https://ifr.org/free-downloads>)

(IFR)



↓
↓
(OUR FOCUS)
↓

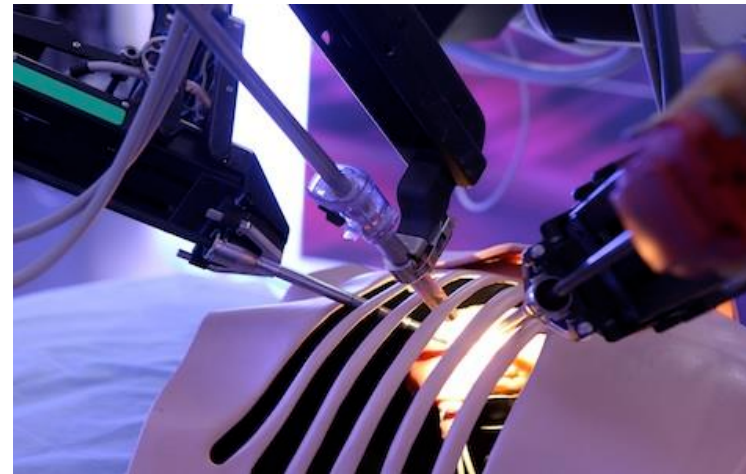
VS

Implies
moving
between
places...
they need
autonomous
navigation



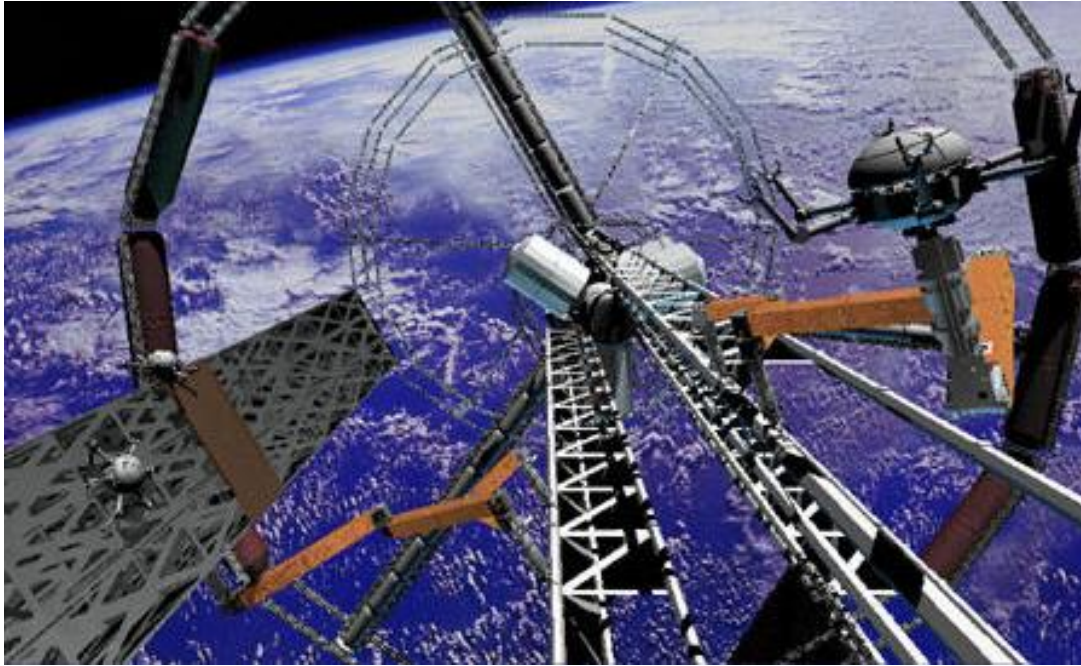
- ✓ A robot system is a system comprising robot(s), end-effector(s) and any machinery, equipment, or sensors supporting the robot performing its task.
- ✓ According to the definition, "a degree of autonomy" is required for service robots ranging from partial autonomy (including human robot interaction) to full autonomy (without active human robot intervention). In this context human robot-interaction means information and action exchanges between human and robot to perform a task by means of a user interface.

Medical robots



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- ✓ Manipulating industrial robots (which can be either fixed in place or mobile) could also be regarded as service robots, provided they are installed in non-manufacturing operations. Service robots may or may not be equipped with an arm structure as is case with some industrial robots. Often, but not always, service robots are mobile.

Space robots



Autonomous vehicles



Traffic Ahead

Many carmakers are developing prototype vehicles that are capable of driving autonomously in certain situations. The technology is likely to hit the road around 2020.



BMW



Mercedes-Benz



Nissan



Google



General Motors

VEHICLE	5 Series (modified)	S 500 Intelligent Drive Research Vehicle	Leaf EV (modified)	Prius and Lexus (modified)	Cadillac SRX (modified)
KEY TECHNOLOGIES	• Video camera tracks lane markings and reads road signs • Radar sensors detect objects ahead • Side laser scanners • Ultrasonic sensors • Differential GPS • Very accurate map	• Stereo camera sees objects ahead in 3-D • Additional cameras read road signs and detect traffic lights • Short- and long-range radar • Infrared camera • Ultrasonic sensors	• Front and side radar • Camera • Front, rear, and side laser scanners • Four wide-angle cameras show the driver the car's surroundings	• LIDAR on the roof detects objects around the car in 3-D • Camera helps detect objects • Front and side radar • Inertial measuring unit tracks position • Wheel encoder tracks movement • Very accurate map	• Several laser sensors • Radar • Differential GPS • Cameras • Very accurate map

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- ✓ In some cases, service robots consist of a mobile platform on which one or several arms are attached and controlled in the same mode as the arms of industrial robot. Furthermore, contrary to their industrial counterparts, service robots do not have to be fully automatic or autonomous. In many cases these machines may even assist a human user or be tele-operated.

Teleoperated and telepresence robots



What makes an **autonomous robot**?

↓ architectures include sensor / planning / acting

A machine gets information from a set of sensors and upon these accomplish its task autonomously by moving its body parts ...

decide what
to do

Plan

Sense

Act

to perceive
environment

implement
action

(Plan-sense-
act
architecture)

always
on autom.
Robots

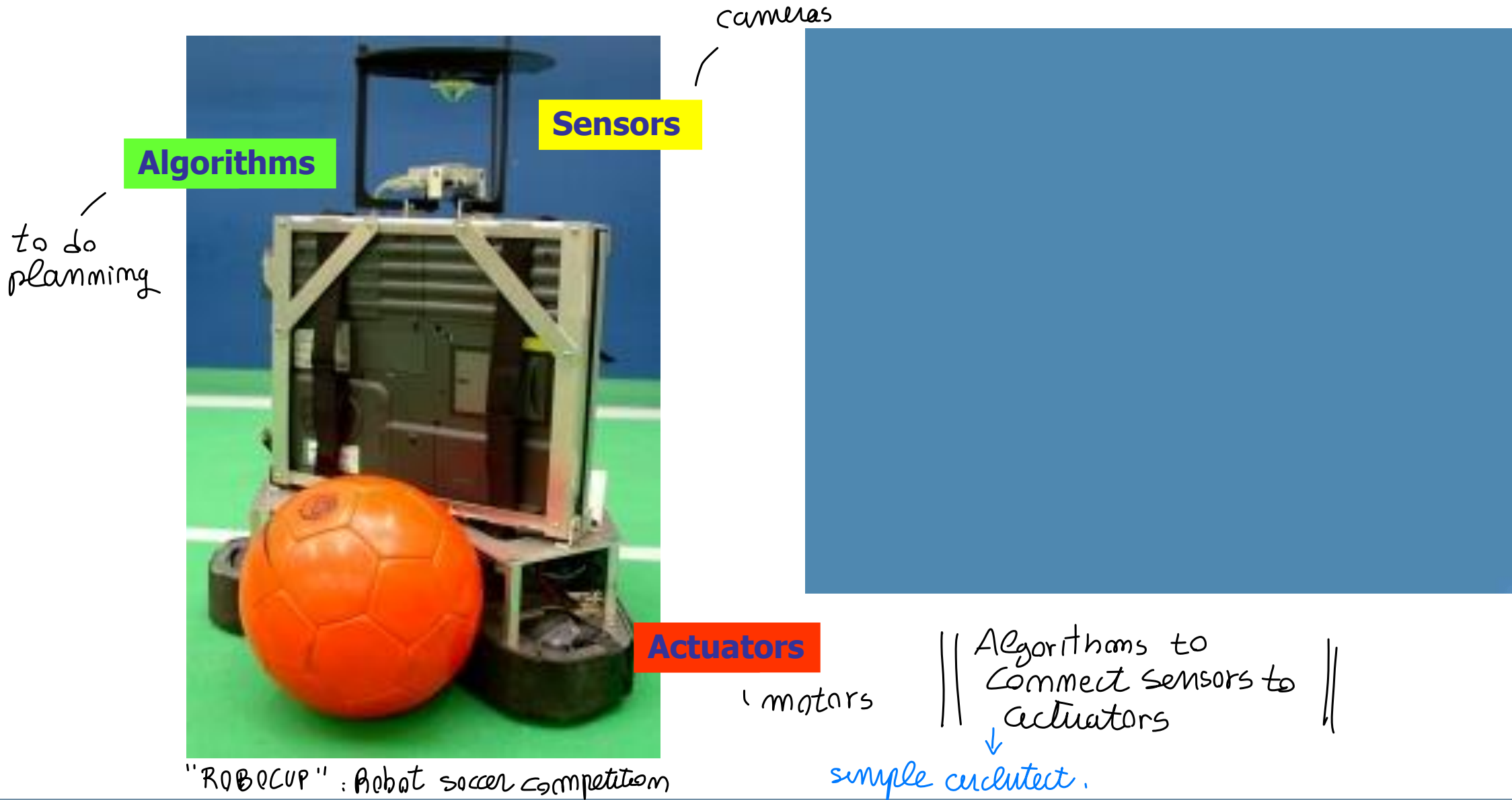


Note: The **Sense-Plan-Act** model is just one possible cognitive architecture for autonomous robots (Cognitive Robotics)

There
are other
possible
PARALLEL
ARCH.



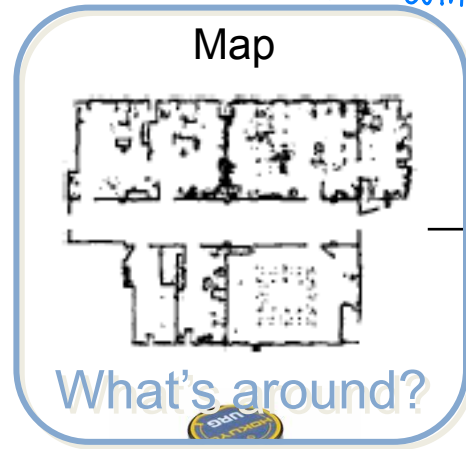
What does it make a mobile robot?



A Simplified Sense-Plan-Act Architecture

(REFERENCE ARCH.)
 → how to control kinematics
 → how to tell where it is
 → how build a map of env
 → plan traj / map

also free space of env. must be known



trajectory comes from this Alg.

Lower Frequency

Trajectory Planning

$f = 1 \text{ Vsec}$

to plan traj goes need to know destination & current

Goal Position



High Frequency

Trajectory (reference)

faster freq.

Trajectory Following
(and Obstacle Avoidance)



Current Position

Need to know where

Where am I?

Mapping



Sensors to detect obstacle

how to build a map of environment

Motion Commands
(Wheel Speed)

kinematics,
how commands became motion



(Reference architecture)
(studied piece by piece)

Localization

how to estimate position

